Eduardo **Rodríguez Sánchez**

COMPUTER ENGINEER · HPC MASTER STUDENT

Madrid, Spain

rod.san.eduardo@gmail.com | https://erodriguez.dev/ | nekronos-gh | eduardo-rodriguez-sanchez

Education

Univ. of Luxembourg / Sofia Univ. St. Kliment Ohridski *Belval / Sofia*

M.S. IN HIGH PERFORMANCE COMPUTING — EUMASTER4HPC DOUBLE DEGREE *Sep. 2025 - Present*

- Focus: parallel algorithms, HPC system architecture, accelerator programming, and performance analysis.
  GPA: 17.5/20.

UC3M / Georgia Institute of Technology *Madrid / Georgia, USA*

B.S. IN COMPUTER ENGINEERING (EXCHANGE YEAR: GEORGIA TECH) *Sep. 2020 - Jun. 2024*

- Coursework: Operating Systems, Networks, Distributed Systems, Machine Learning, Cryptography.
  GPA: 9.1/10 (UC3M) · 4.0/4.0 (Georgia Tech).
  Thesis: C++ load-balancing proxy sustaining 86 Gb/s of trace forwarding across millions of concurrent connections.

Skills

Programming C, C++, Python
HPC / Systems OpenMP, MPI, AVX2, Multithreading, Profiling, Linux, GDB, Valgrind
Distributed / Cloud Kubernetes, Spark, 5G Networking, gRPC, Prometheus, Grafana
Languages Spanish (native), English (fluent)

Experience

Ericsson *Madrid, Spain*

SOFTWARE DEVELOPER *Dec. 2024 - Aug. 2025*

- Shipped User Plane Analytics features and antenna geo-redundancy for the 5G Core, and cut multi-threaded lock contention to resolve critical production issues.
- Led the migration of the 5G RIB codebase to the Madrid site and redesigned intra-forwarding-module communication to run entirely inside Kubernetes, eliminating external network hops.

Samsung Electronics (Zhilabs) *Madrid, Spain*

SOFTWARE ENGINEER *Jun. 2023 - Oct. 2024*

- Built a network trace proxy sustaining 86 Gb/s with a recursive-descent binary parser for trace analysis, and parallelized a multi-antenna traffic simulator to speed up testing.
- Designed a RAN Assistant, a proof-of-concept multi-agent LLM/NLP/ML system for telecom Q&A, anomaly detection, and automated root-cause analysis.

Research

SnT (Security, Reliability and Trust) *Luxembourg*

RESEARCH ASSISTANT *Oct. 2025 - Apr. 2026*

- Prototyped a Byzantine fault-tolerant PBFT consensus engine in C++ on an async networking stack, including a cross-replica-group coordination protocol.
- Built failure-injection scenarios and reproducible benchmarking tooling to validate consensus performance under adversarial conditions.

Projects

Rule110 Fast

GITHUB.COM/NEKRONOS-GH/RULE110

Cellular-automaton simulator vectorized with AVX2 and parallelized with OpenMP, using aligned memory in modern C++23.

TinyFile

GITHUB.COM/NEKRONOS-GH/TINYFILE

Zero-copy file-compression service built on POSIX message queues and shared memory, driven by a ring-buffer state machine.

PBFT

GITHUB.COM/NEKRONOS-GH/PBFT-CPP

Full 3-phase consensus implementation in C++ (Salticidae) with view changes and checkpointing, instrumented via Prometheus/Grafana.